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Course: CSE350

Section: 10

Assignment: 01

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Ans to the q. no-1

here,

$$V_{\text{ref}} = -10\text{V}$$

$$R_{\text{ref}} = 10\text{K}\Omega$$

$$R = 5\text{K}\Omega$$

[a]

$$I_1 = \frac{V_{\text{ref}}}{2R} = \frac{-10}{2 \times 5} = -1\text{mA}$$

$$I_2 = \frac{I_1}{2} = \frac{-1}{2} = -0.5\text{mA}$$

$$I_3 = \frac{I_2}{2} = \frac{-0.5}{2} = -0.25\text{mA}$$

Ans

[b]

$$\text{resolution} = \text{step size} = 1 \text{ LSB}$$

$$1 \text{ LSB} = 001$$

$$\therefore V_o = -V_{\text{ref}} \times \frac{R_f}{2R} \times \left[b_1 + \frac{b_2}{2} + \frac{b_3}{4} \right]$$

$$= -(-10) \times \frac{10}{2 \times 5} \left[0 + \frac{0}{2} + \frac{1}{4} \right]$$

$$= 10 \times \frac{1}{4}$$

$$= 2.5\text{V}$$

Ans

[c]

for maximum output,

$$\text{Input} = 111$$

$$\begin{aligned}\therefore V_{\max} &= -V_{\text{ref}} \times \frac{R_f}{2R} \times \left[b_1 + \frac{b_2}{2} + \frac{b_3}{4} \right] \\ &= -(-10) \times \frac{10}{2 \times 5} \left[1 + \frac{1}{2} + \frac{1}{4} \right] \\ &= 10 \times \frac{7}{4} = 17.5\end{aligned}$$

for minimum output

$$\text{Input} = 000$$

$$\begin{aligned}\therefore V_{\min} &= -V_{\text{ref}} \times \frac{R_f}{2R} \times \left[b_1 + \frac{b_2}{2} + \frac{b_3}{4} \right] \\ &= -(-10) \times \frac{10}{2 \times 5} \times \left[0 + \frac{0}{2} + \frac{0}{4} \right] \\ &= 0V\end{aligned}$$

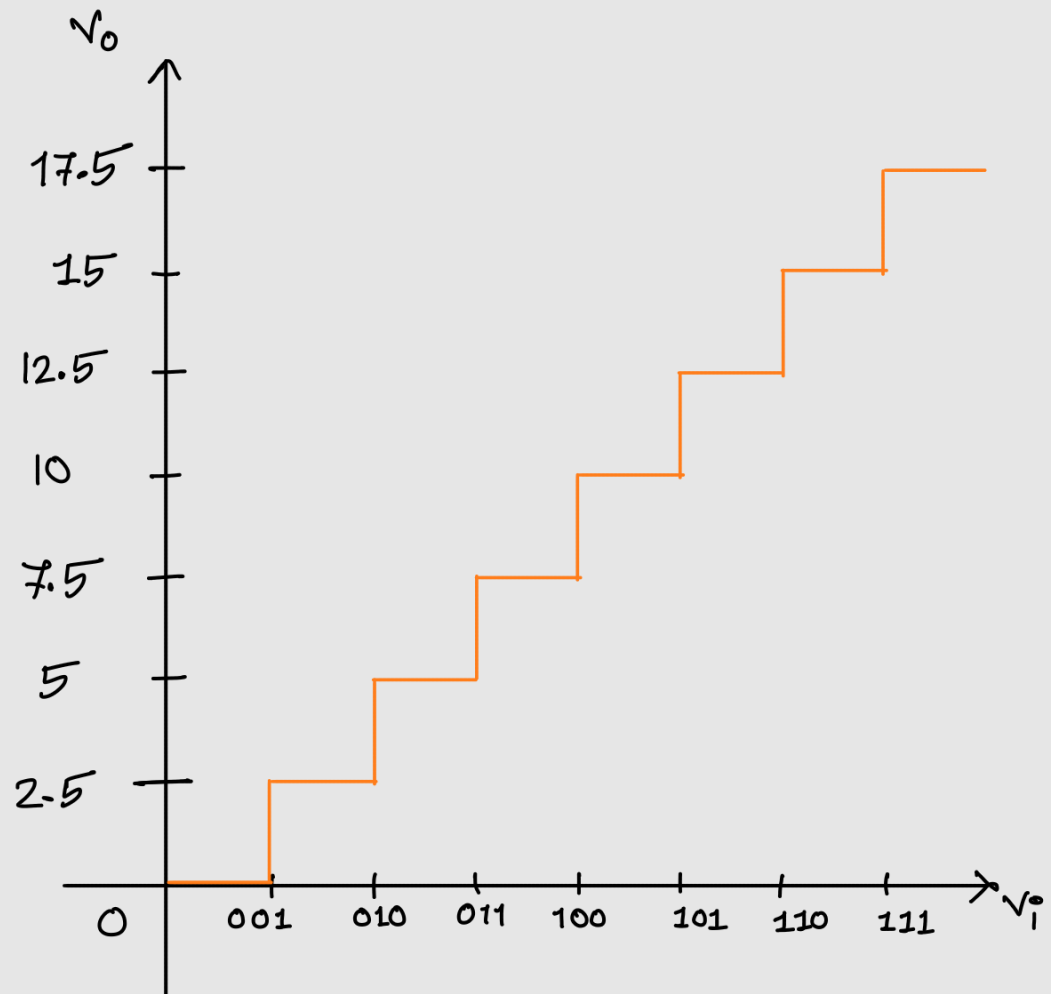
[d]

$$1 \text{ MSB} = 100$$

$$\begin{aligned}\therefore V_0 &= -V_{\text{ref}} \times \frac{R_f}{2R} \times \left[b_1 + \frac{b_2}{2} + \frac{b_3}{4} \right] \\ &= -(-10) \times \frac{10}{2 \times 5} \times \left[1 + \frac{0}{2} + \frac{0}{4} \right] \\ &= 10V\end{aligned}$$

$$\therefore \frac{1}{2} \text{ MSB} = \frac{10}{2} = 5V. \quad \underline{\text{Ans}}$$

e



Ans. to the q.no-2

9

here,

$$V_{ref} = +8V$$

$$R_3 = R_f$$

$$\begin{aligned} V_o &= V_{ref} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right) \times \frac{R_f}{R_3} \\ &= +8 \times \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} \right) \quad [b_4 = 0] \end{aligned}$$

Binary Input	V_o
0 0 0 0	0
0 0 1 0	2
0 1 0 0	4
0 1 1 0	6
1 0 0 0	8
1 0 1 0	10
1 1 0 0	12
1 1 1 0	14

10

max output (for input 1111)

$$\begin{aligned} V_{max} &= V_{ref} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right) \times \frac{R_f}{R_3} \\ &= 8 \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) \\ &= 15 \end{aligned}$$

LSB (for input 0001)

$$\begin{aligned}V_{\min} &= V_{\text{ref}} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right) \frac{R_f}{R_3} \\&= 8 \left(0 + \frac{0}{2} + \frac{0}{4} + \frac{1}{8} \right) \\&= 8 \times \frac{1}{8} = 1V.\end{aligned}$$

$$\therefore \text{Ratio } V_{\max} \text{ and LSB voltage} \Rightarrow \frac{V_{\max}}{V_{\text{LSB}}} = \frac{15}{1} = 15$$

Ans

②

here, output = 1010

$$V_o = 5V$$

$$V_o = V_{\text{ref}} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right) \times \frac{R_f}{R_3}$$

$$\Rightarrow 5 = 8 \left(1 + \frac{0}{2} + \frac{1}{4} + \frac{0}{8} \right) \times \frac{R_f}{R_3}$$

$$\Rightarrow 5 = 10 \times \frac{R_f}{R_3}$$

$$\Rightarrow \frac{R_3}{R_f} = \frac{10}{5}$$

$$\Rightarrow \frac{R_3}{R_f} = 2$$

Ratio of $R_3 : R_f$ should be 2:1.

Ans. to the q.no-3

a

here,

$$V_{ref} = 10V$$

$$R_1 = 5k\Omega, R_2 = 10k\Omega, R_3 = 10k\Omega, R_4 = 15k\Omega$$

$$R_{total} = 5 + 10 + 10 + 15 = 40k\Omega.$$

$$V_1 = 0V$$

$$V_2 = \frac{R_1}{R_{total}} \times V_{ref} = \frac{5}{40} \times 10 = 1.25V$$

$$V_3 = \frac{R_1 + R_2}{R_{total}} \times V_{ref} = \frac{5 + 10}{40} \times 10 = 3.75V$$

$$V_4 = \frac{R_1 + R_2 + R_3}{R_{total}} \times V_{ref} = \frac{5 + 10 + 10}{40} \times 10 = 6.25V$$

$$V_{max} = 10V$$

Quantization Range	Digital Signal	LED1	LED2
6.25 - 10	11	ON	ON
3.75 - 6.25	10	ON	OFF
1.25 - 3.75	01	OFF	ON
0 - 1.25	00	OFF	OFF

b

Sample value (V)	time (s)	output	LED1	LED2
5	0.0	10	ON	OFF
7.4	0.5	11	ON	ON
5.71	3.0	10	ON	OFF
1.22	4.0	00	OFF	OFF

Ans. to the q. no-4

here,

$$V_{ref} = 5V$$

$$N = 8$$

a

$$\text{Voltage range} = (0 - V_{ref}) = 5V$$

$$\text{number of steps} = 2^N = 2^8 = 256$$

$$\begin{aligned} \text{step size} &= \frac{\text{voltage range}}{\text{number of steps}} \\ &= \frac{5}{256} = 0.0195V \end{aligned}$$

b

here, $n = 200$.

Now

$$\frac{V_A}{V_{ref}} = \frac{n}{2^N}$$

$$\Rightarrow \frac{V_A}{5} = \frac{200}{2^8}$$

$$\Rightarrow V_A = \frac{200}{2^8} \times 5 = 3.9V.$$

Am

C

first integration period:

$$t_1 = \frac{2^N}{\text{clock rate}} = \frac{2^8}{1 \times 10^6} \\ = 2.56 \times 10^{-4} \text{ s}$$

2nd integration period:

$$t_2 = \frac{m}{\text{clock rate}} = \frac{200}{1 \times 10^6} \\ = 2 \times 10^{-4} \text{ s}$$

$$\therefore \text{total conversion time} = t_1 + t_2 \\ = 2.56 \times 10^{-4} + 2 \times 10^{-4} \\ = 4.56 \times 10^{-4} \text{ s}$$

D

for worst sampling rate,

$$t_1 = t_2 = 2.56 \times 10^{-4} \text{ s}$$

$\therefore$ worst case conversion duration,

$$t_1 + t_2 = 2.56 \times 10^{-4} + 2.56 \times 10^{-4} \\ = 5.12 \times 10^{-4} \text{ s}$$

$$\therefore \text{max sampling rate} = \frac{1}{5.12 \times 10^{-4}}$$

$$= 1953.125 \text{ Hz.}$$

Ans

[e]

for 4x sampling rate,

$$\text{max count should be } \Rightarrow \frac{256}{4} = 64$$

$\therefore$ The counter should have $\log_2(64) = 6$ bits

Ans